

Education

- 2026 **The University of Texas at Austin** | *Ph.D. in Computer Science*
Advisor: [Prof. Aditya Akella](#)
- 2020–2021 **University of Wisconsin-Madison** | *Ph.D. in Computer Science*
(Transferred) Advisor: [Prof. Aditya Akella](#)
- 2020 **Indian Institute of Technology, Delhi** | *B.Tech. in Computer Science and Engineering*

Research Direction

Computer systems must meet stringent *latency* (e.g., sub-second request completion) and *resource* (e.g., scaling to meet request bursts) requirements under dynamic conditions. This demands *adaptive and robust policies* across the stack. My research combines ideas from *Machine Learning (ML)*, *Optimization*, *Programming Languages*, and *Formal Methods* into computer systems to tackle these challenges through two complementary directions:

- *Abstractions and controllers for modern applications*: I have devised novel *abstractions*, intuitive *interfaces* and optimal *controllers* to address challenges in *communication* [7, 16, 4], *resource management* [2, 6, 9] and *production deployments* [11] of microservice and serverless applications.
- *ML for the OS*: I have developed a range of *ML-driven policies* – from *neural networks for critical-path decisions* [3] to *encoders* [12], and *LLMs* [13, 1] for various OS policies, along with techniques to *robustify* their operations using analytical models [2,10], formal verification [3], and monitoring frameworks [14].

Publications

Peer-Reviewed Conference Publications

- EuroSys '27 *Vulcan: Instance-specialized, Verifiable Systems Heuristics Through LLM-driven Search*. [1]
[Rohit Dwivedula](#), [Divyanshu Saxena](#), [Sujay Yadalam](#), [Eric Hayden Campbell](#), [Daehyeok Kim](#), and [Aditya Akella](#).
- NSDI '26 *Towards Performance Robustness for Microservices*. [2]
[Divyanshu Saxena](#), [Gaurav Vipat](#), [Jiaxin Lin](#), [Jingbo Wang](#), [Isil Dillig](#), [Sanjay Shakkottai](#), and [Aditya Akella](#).
- EuroSys '26 *Canopy: Property-Driven Learning for Congestion Control*. [3]
[Chenxi Yang](#), [Divyanshu Saxena*](#), [Rohit Dwivedula*](#), [Kshiteej Mahajan](#), [Swarat Chaudhuri](#), and [Aditya Akella](#).
- PLDI '26 *MatchBox: A Semantic Foundation for Data Plane Portability*. [4]
[Eric Hayden Campbell](#), [Robert Zhang](#), [Divyanshu Saxena](#), [Aditya Akella](#), and [Isil Dillig](#).
🏆 *Distinguished Paper Award*.
- ASPLOS '26 *Enabling fast networking in the public cloud*. [5]
[Alireza Sanaee](#), [Vahab Jabrayilov](#), [Ilias Marinos](#), [Farbod Shahinfar](#), [Divyanshu Saxena](#), [Kostis Kaffes](#), and [Gianni Antichi](#).
- ICLR '25 *CONGO: Compressive Online Gradient Optimization*. [6]
[Jeremy Carleton](#), [Prathik Vijaykumar](#), [Divyanshu Saxena](#), [Dheeraj Narasimha](#), [Srinivas Shakkottai](#), and [Aditya Akella](#).
- ASPLOS '25 *Copper and Wire: Bridging Expressiveness and Performance for Service Mesh Policies*. [7]
[Divyanshu Saxena](#), [William Zhang](#), [Shankara Pailoor](#), [Isil Dillig](#), and [Aditya Akella](#).
- SoCC '23 *Yama: Providing Performance Isolation for Black-Box Offloads*. [8]
[Tao Ji](#), [Divyanshu Saxena](#), [Brent E. Stephens](#), and [Aditya Akella](#)

EuroSys '22 [Memory Deduplication for Serverless Computing with Medes.](#) [9]
[Divyanshu Saxena](#), Tao Ji, Arjun Singhvi, Junaid Khalid, and Aditya Akella.

Peer-Reviewed Workshop Papers

DIMES at SOSP '26 [TierSizer: Counterfactual Reasoning for DRAM Sizing in Tiered Memory Systems.](#) [10]
Sujay Yadalam, Saarth Deshpande, [Divyanshu Saxena](#), Venkat Arun, Neeraja Yadwadkar, Aditya Akella.

LLM4Code at ICSE '26 [Continuous Benchmark Generation for Evaluating Enterprise-scale LLM Agents.](#) [11]
[Divyanshu Saxena](#), Rishikesh Maurya, Xiaoxuan Ou, Gagan Somashekar, Shachee Mishra Gupta, Arun Iyer, Yu Kang, Chetan Bansal, Aditya Akella, and Saravan Rajmohan.

MLForSys at NeurIPS '25 [A Joint Learning Approach to Hardware Caching and Prefetching.](#) [12]
Samuel Yuan, [Divyanshu Saxena](#), Jiayi Chen, Nihal Sharma, and Aditya Akella.

HotNets '25 [Man-Made Heuristics Are Dead. Long Live Code Generators!](#) [13]
Rohit Dwivedula, [Divyanshu Saxena](#), Aditya Akella, Swarat Chaudhuri, and Daehyeok Kim.

HotOS '25 [How I learned to stop worrying and love learned OS policies.](#) [14]
[Divyanshu Saxena*](#), Jiayi Chen*, Sujay Yadalam, Yeonju Ro, Rohit Dwivedula, Aditya Akella, Christopher Rossbach, and Mike Swift.

MLForSys at NeurIPS '23 [On a Foundation Model for Operating Systems.](#) [15]
[Divyanshu Saxena](#), Nihal Sharma, Donghyun Kim, Rohit Dwivedula, Jiayi Chen, Chenxi Yang, Sriram Ravula, Zichao Hu, Aditya Akella, Sebastian Angel, Joydeep Biswas, Swarat Chaudhuri, Isil Dillig, Alex Dimakis, P. Brighten Godfrey, Daehyeok Kim, Chris Rossbach, and Gang Wang.

ApPLIED at PODC '23 [Invited Paper: Towards Efficient Microservice Communication.](#) [16]
[Divyanshu Saxena](#), William Zhang, Madhav Tummala, Saksham Goel, and Aditya Akella.

Work Experience

Aug 2026-Present **University of California, Berkeley** | *Postdoctoral Scholar at NetSys Lab, with [Prof. Rishabh Iyer](#)*

Summer 2025 **M365 Research** | *Research Intern with Systems Innovation Group*

Summer 2022 **Microsoft Research** | *Research Intern at Networking Research Group*

Scholastic Achievements

2022, 2025 Student Travel Grants to attend EuroSys 2022, ASPLOS 2025, and HotOS 2025.

2023, 2025 Professional Development Award at UT Austin for Summer 2023, Spring 2025.

2020 Departmental Scholarship of USD 3000 for the academic session 2020-21 at UW-Madison.

2016 All India Rank 64 in Joint Entrance Exam (Advanced) among 1.5 million applicants.

2015 All India Rank 61 in Kishore Vaigyanik Protasahan Yojana (KVPY) conducted by IISc Bangalore.

2017 Felicitated with "Design Innovation Summer Award" – 2017 by the Ministry of Human Resource Development (MHRD), given to selected projects from IIT Delhi, for the project *Person Counter and Display Device*

2016 Qualified the National Standard Examination in Physics (NSEP) and Chemistry (NSEC).

2012 Conferred a nine-year scholarship on qualifying National Talent Search Examination (2012), conducted by NCERT.

Teaching Experience

Fall 2023 Mentor | Directed Reading Program (DiRP) on "Emerging Cloud Infrastructure"

Fall 2020 Teaching Assistant | Programming III, at UW-Madison

Fall 2019 Teaching Assistant | Artificial Intelligence, at IIT Delhi

Service

Program Committee | SoCC 2025, 2026; MemNetAI Workshop at SIGCOMM 2026

Reviewer | Journal for Systems Research 2025; ML for Systems Workshop at NeurIPS 2025, 2026

Student Volunteer | HotNets 2022 hosted in Austin

Artifact Evaluation Committee | SOSP 2023; OSDI 2022; ATC 2022

Mentorship

2022-23 William Zhang | B.S., UT Austin → Google

Project: A new language for service mesh policies

2024-25 Samuel Yuan | B.S., UT Austin → OpenAI

Project: Joint learning for hardware caching and prefetching

2024-25 Ebaad Imran | B.S., UT Austin → Optiver

Project: Joint autoscaling and admission control for microservices

2025-26 William Wang | B.S., UT Austin

Project: Implementing guardrails for hugepage management

2025-26 Shelby Yang | B.S., UT Austin

Project: Building robust memory tiering systems

2024-26 Gaurav Vipat | Pre-doctoral Researcher, UT Austin → Ph.D. Student, UT Austin

Project: Tuning NUMA configuration knobs for improved performance

2025-Ongoing Diego Wearden | B.S. + M.S., UT Austin

Project: Exploring vulnerabilities with learned CPU load balancing in the OS kernel

2025-Ongoing Shruti Narwaney | B.S., UT Austin

Project: Identifying out-of-distribution inputs for flow size predictors

2025-Ongoing Kailer Laino | B.S., UT Austin

Project: Agentic approaches to congestion control